Appendix: UX Research & Evaluation

Part 1 - User Research Summary

Methodology

We conducted **14 semi-structured interviews** (45–60 minutes each) with personas spanning the enterprise buying committee that any GAP deal will pass through: compliance officers, security and legal gatekeepers, platform engineering deployers, executive sponsors, and frontline operations end users. Persona agents are included in the prototype repository. Interviews covered current pain points, walkthrough of the GAP press release language, prototype demo at *claude-gap-prototype.watershed.technology*, and willingness to pilot. **Bias to flag:** participants were recruited from AI-curious networks; AI-skeptical leaders are likely underrepresented.

Personas Interviewed

#	Persona	Role & Org	Segment
P01	Maria Chen	Head of AI Governance, Top-5 US Bank	Primary buyer — banking compliance
P02	Dr. Anjali Rao	VP Clinical Ops, F500 Pharma	Adjacent segment — pharma/life sciences
P03	Marcus Bauer	CISO, Mid-size Investment Bank	Security veto holder
P04	Priya Shankar	Principal Platform Engineer, Insurer	IT integrator/deployer
P05	James Whitfield	Chief AI Officer, Asset Manager ($1.2T)	Executive sponsor
P06	Sofia Reyes	Sr. PM Enterprise AI, Healthcare SaaS	Builder on top of Claude
P07	Eric Tanaka	Director Claims Ops, P&C Insurer	Operations end customer
P08	Lin Zhang	Controller, Multinational Manufacturer	Finance / SOX owner
P09	Robert Mendelsohn	General Counsel, Health System	Legal & risk veto
P10	Devon Park	IT Admin / IAM Lead, Regional Bank	Day-to-day operator
P11	Kelechi Okafor	Sr. Internal Auditor, Tier-1 Bank (London)	Second-line audit
P12	Helena Voss	Data Protection Officer, EU Retail Bank	GDPR / EU AI Act lead
P13	Tom Heller	Eng Manager, In-house AI Platform	Skeptic / build-vs-buy
P14	Naomi Goldberg	Sr. Sanctions Analyst	Frontline approver / UX end user

Positives

Finding 1. The tamper-evident audit log is GAP's emotional anchor. Ten of 14 personas surfaced the cryptographic audit log as their first point of enthusiasm, without prompting. Maria (P01, compliance): "The tamper-evident log is the single thing I've been asking three vendors for." Kelechi (P11, audit): "I can trust a 12-month-old record without verifying with the engineering team that no one tampered with it." Robert (P09, GC): "Something I can hand to outside counsel during an investigation."

Finding 2. The deny path validates as a v1 design decision. Three personas independently described deny + reason capture as half the workflow. Naomi (P14, frontline approver): "Deny with a clean reason that survives audit review three months later is what I get paid for." Robert (P09) called denied actions "a defensible record" in litigation. Kelechi (P11) said deny capture is "where I find the interesting audit findings."

Finding 3. Plain-text policy is the unlock for non-engineering buyers. Compliance and finance personas (P01, P08, P09) were emphatic that their teams will not learn DSLs but will read a YAML or plain-text file with comments. Priya (P04) confirmed the platform-engineering side: "OPA underneath — my security team already uses it for Kubernetes admission control."

Concerns

Finding 4. Policy authoring is the central skepticism point. Every persona who reviewed the PR-FAQ flagged that the prototype's single hardcoded rule is a long way from a production authoring experience. Maria (P01): "I need to see the policy authoring experience end-to-end before I sign anything." This is the gap between the press release language and what the team has demonstrated.

Finding 5. Latency is a silent killer. Priya (P04), Marcus (P03), and Eric (P07) all flagged "minimal latency" as too vague. The team needs concrete p50 and p99 numbers under load before pilot conversations get serious.

Finding 6. Indemnification will dominate contract negotiation. Robert (P09, GC), Marcus (P03, CISO), and James (P05, CAO) all flagged that the External FAQ #10 language ("Anthropic is not liable for the underlying action") is the wrong opening position. A tiered indemnification posture is needed before the first pilot contract.

Finding 7. Banking-first framing leaves adjacent segments behind. Anjali (P02, pharma), Helena (P12, EU), and Marcus (P03, CISO) noted the launch materials read as banking-first. Anjali was sharpest: "I do not want to be the pharma example that proves the product was not built for us." Pharma needs 21 CFR Part 11 e-signature commitments; the EU needs EU AI Act Article 14 alignment documentation and an EU-resident signing-key option.

Finding 8. Workflow integration matters more than features. Lin (P08) needs Workiva GRC integration. Devon (P10) needs SCIM provisioning and approver-timeout escalation. Sofia (P06) needs an OEM/embed story to white-label for her hospital customers. A standalone GAP without these does not earn budget.

Pilot Disposition

Strong yes (would pilot in v1): Maria (P01), Priya (P04), James (P05), Lin (P08). **Conditional yes:** Sofia (P06, needs OEM story), Helena (P12, needs EU AI Act documentation), Devon (P10, needs SCIM). **Not in v1:** Anjali (P02, awaiting Part 11 e-signature), Tom (P13, build-vs-buy skeptic). **Gatekeepers (no veto but heavy contract negotiation):** Marcus (P03), Robert (P09), Kelechi (P11).

Part 2 - Evaluation & Experiment Design

Approach

GAP combines a deterministic policy proxy with a non-deterministic **Supervisor Agent** (Claude 4.7 Sonnet) that classifies intercepted tool calls as Safe, Ambiguous, or Dangerous. The policy proxy is testable by inspection; the Supervisor Agent's reasoning is not. We designed two complementary validation efforts: an **offline evaluation** against a Golden Dataset of expert-labeled cases to certify the Supervisor's reasoning quality, and an **online A/B experiment** against the human-bottlenecked baseline to measure whether Agentic Supervision actually unlocks velocity in production.

Eval Set: Supervisor Agent Reasoning Benchmark

Golden Dataset: 500 prompts labeled Deny / Escalate / Approve by a panel of three Enterprise Security Experts, balanced to stress-test different failure modes.

Case type	N	Description	What it validates
Compliant	300	Routine wire transfers under limit to known vendors	False negative rate (unnecessary escalation)
Ambiguous	100	Technically compliant but contextually suspicious (unusual time, first-time vendor)	Reasoning quality on edge cases
Adversarial	100	Intentional breaches: credential theft, jailbreak reasoning to bypass limits	False positive rate (dangerous approval) — primary gate

Grading strategy: Claude 4.7 Opus serves as Agent-as-Judge for the Sonnet Supervisor-Candidate, since recruiting a 3-expert panel for every model iteration does not scale. Human expert panel is reserved for final release-candidate validation. **Judge bias risk:** Opus may grade Sonnet too leniently; we mitigate with a held-out human-validated subset on every release.

Eval Success Targets

Metric	Target	Why
Correctness vs. Expert Panel	> 99.5% overall	Alignment with ground-truth labels. Greater than 99% target on Adversarial alone.
Adversarial False Approval Rate	0%	Zero is the launch gate. Any Approve on a labeled-Dangerous case blocks release. This is the metric that wins CISO trust.
Hallucination Rate	< 1%	% of justifications referencing non-existent policies or wrong tool parameters.
Policy Attribution Accuracy	100%	Every Deny or Escalate must correctly cite the policy rule violated. Misattribution = Critical Reasoning Failure.
Latency (P50 / P95)	< 1.5s / < 3s	Supervision must feel instantaneous to preserve agent velocity.
Cost per supervision event	< $0.02	Sonnet enterprise pricing, April 2026. Unit economics

		gate.

Online Experiment: Agentic Supervision vs. Human Bottleneck

21-day A/B test. Control: every high-risk tool call goes to a human queue. Treatment: the Supervisor Agent auto-approves cases that match deterministic policy with greater than 98% confidence; only Ambiguous or extreme-risk cases escalate. **Target metric:** Mean Time to Action (MTTA), with a launch criterion of > 70% reduction for low-complexity calls. **Guardrails:** Downstream Incident Rate within −1% of human baseline; Deflection Rate (auto-resolved share) > 50% to justify supervisor inference cost.

How User Research Validates the Eval Design

Three independent research signals confirm the eval design measures what enterprise buyers actually care about: **(1)** Marcus (P03, CISO) said his entire pilot approval hinges on "a written security architecture review." The Adversarial 0% False Approval target is the technical proof that backs that review. **(2)** Naomi (P14, frontline approver) and Eric (P07, ops director) both said GAP only delivers value if the auto-approve path handles routine cases; otherwise "I haven't gained anything." The Deflection Rate guardrail (>50%) directly tests this. **(3)** Priya (P04, platform engineer) demanded concrete latency numbers; the P50 < 1.5s / P95 < 3s targets give her the answer she needs before signing.

Risks & Limitations of the Validation Approach

- **Model drift.** If the underlying Claude 4.7 model updates, the Supervisor's calibration for ambiguity may shift, spiking human escalations. Re-run the full Golden Dataset on every model version.
- **Judge leniency.** Opus-as-Judge may overstate Sonnet readiness. Held-out human panel sampling on every release candidate is the mitigation.
- **Silent failure risk.** If Primary and Supervisor agents share a reasoning bias, the Supervisor may hallucinate a dangerous action as compliant. The chain-of-thought verification middleware (forbidden-pattern detection) is the defense.
- **Supervisor fatigue.** If escalation false-alarm rate is too high, human approvers begin ignoring flags. Tracked via the Compliant-case false-positive rate in the Golden Dataset.

Implications for Product Direction

- **Ship the policy starter library as a launch-blocker, not a nice-to-have** (Finding 4).
- **Add a structured deny-reason taxonomy in the policy file** (Finding 2; Naomi P14's specific UX worry).
- **Publish concrete latency numbers before pilot conversations** (Finding 5; the eval's P50 < 1.5s / P95 < 3s targets give the answer to Priya P04 and Marcus P03).
- **Pre-prep a tiered indemnification posture** before first pilot contract (Finding 6).
- **Hold the 0% Adversarial False Approval line as the public proof point** for CISO conversations — Marcus (P03) explicitly said this is what would unblock his sign-off.
- **Develop segment-specific launch materials** for pharma (21 CFR Part 11) and EU (AI Act Article 14) before the second wave of GTM (Finding 7).